SCIENTIFIC COMPUTING

Comparison of Jacobi and Gauss-Seidel Iterative Methods for the Solution of Systems of Linear Equations

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Linear Systems • Convergence Analysis

JACOBI VS SEIDEL ITERATIONS

Core Loop & Differences

Jacobi Method

Calculations depend strictly on previous iteration values.

```
sum += A[i][j] * x_old[j];
 x_new[i] = (d[i] - sum) / A[i][i];
 // buffer swap after full pass
 x_old = x_new;
```

Gauss-Seidel Method

In-place substitution immediately uses newly computed unknowns.

```
sum += A[i][j] * x_curr[j];
 x_curr[i] = (d[i] - sum) / A[i][i];
 // immediately available for row i+1
```

TEST CASES

System Inputs and Computed Solutions

Test Case 0

3×3 System

Order: n = 3

```
x1 = 1.000
 x2 = 2.000
 x3 = -1.000
```

Test Case 1

2×2 System

Order: n = 2

```
x1 = 1.000
 x2 = 1.000
```

Test Case 2

4×4 System

Order: n = 4

```
x1 = 1.000
 x2 = 1.000
 x3 = 1.000
 x4 = 1.000
```

Test Case 3

5×5 System

Order: n = 5

```
x1 = 1.000
 x2 = 1.000
 x3 = 1.000
 x4 = 1.000
 x5 = 1.000
```

EVALUATION

Iteration Counts and Error Percentages

`Tolerance threshold = 0.0001 (1e-4)`

Test Suite	Order (n)	Jacobi Iterations	Jacobi Error %	Seidel Iterations	Seidel Error %
Test Case 0	n = 3	**8**	`0.000950%`	**4**	`0.000750%`
Test Case 1	n = 2	**8**	`0.004800%`	**5**	`0.000700%`
Test Case 2	n = 4	**11**	`0.001500%`	**5**	`0.000150%`
Test Case 3	n = 5	**12**	`0.001700%`	**5**	`0.000720%`

```
PS C:\Users\Mayank\Desktop\Gauss Jacobi and Seidel Iteration> ./test 0
Running Test Case 0 (n=3)

========== JACOBI ITERATIONS ==========

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Jacobi Iteration 1: x1 = 0.6, x2 = 2, x3 = -1
Gauss-Jacobi Iteration 2: x1 = 1, x2 = 1.96364, x3 = -0.92
Gauss-Jacobi Iteration 3: x1 = 0.980364, x2 = 2.00727, x3 = -1.00364
Gauss-Jacobi Iteration 4: x1 = 1.00145, x2 = 1.99788, x3 = -0.995345
Gauss-Jacobi Iteration 5: x1 = 0.998858, x2 = 2.00056, x3 = -1.0005
Gauss-Jacobi Iteration 6: x1 = 1.00016, x2 = 1.99985, x3 = -0.999716
Gauss-Jacobi Iteration 7: x1 = 0.999928, x2 = 2.00004, x3 = -1.00005
Gauss-Jacobi Iteration 8: x1 = 1.00001, x2 = 1.99999, x3 = -0.999982
Gauss-Jacobi Iteration 9: x1 = 0.999995, x2 = 2, x3 = -1
Gauss-Jacobi Iteration 10: x1 = 1, x2 = 2, x3 = -0.999999
Gauss-Jacobi Iteration 11: x1 = 1, x2 = 2, x3 = -1
Gauss-Jacobi Iteration 12: x1 = 1, x2 = 2, x3 = -1
Gauss-Jacobi Iteration 13: x1 = 1, x2 = 2, x3 = -1
Gauss-Jacobi Iteration 14: x1 = 1, x2 = 2, x3 = -1
Gauss-Jacobi Iteration 15: x1 = 1, x2 = 2, x3 = -1

======= GAUSS-SEIDEL ITERATIONS =======

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Seidel Iteration 1: x1 = 0.6, x2 = 2.05455, x3 = -0.914545
Gauss-Seidel Iteration 2: x1 = 0.988364, x2 = 2.00671, x3 = -0.997002
Gauss-Seidel Iteration 3: x1 = 1.00007, x2 = 2.00028, x3 = -0.999986
Gauss-Seidel Iteration 4: x1 = 1.00003, x2 = 2, x3 = -1
Gauss-Seidel Iteration 5: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 6: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 7: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 8: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 9: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 10: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 11: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 12: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 13: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 14: x1 = 1, x2 = 2, x3 = -1
Gauss-Seidel Iteration 15: x1 = 1, x2 = 2, x3 = -1

========== FINAL COMPARISON ==========

Methods                 No of iteration     Error %          Time (ms)        Status
-------------------------------------------------------------------------------
Jacobi method           8                   0.000950         37.054800        Tolerance reached
Gauss-Seidel method     4                   0.000750         34.030800        Tolerance reached
```

```
PS C:\Users\Mayank\Desktop\Gauss Jacobi and Seidel Iteration> ./test 1
Running Test Case 1 (n=2)

========== JACOBI ITERATIONS ==========

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Jacobi Iteration 1: x1 = 1.25, x2 = 1.33333
Gauss-Jacobi Iteration 2: x1 = 0.916667, x2 = 0.916667
Gauss-Jacobi Iteration 3: x1 = 1.02083, x2 = 1.02778
Gauss-Jacobi Iteration 4: x1 = 0.993056, x2 = 0.993056
Gauss-Jacobi Iteration 5: x1 = 1.00174, x2 = 1.00231
Gauss-Jacobi Iteration 6: x1 = 0.999421, x2 = 0.999421
Gauss-Jacobi Iteration 7: x1 = 1.00014, x2 = 1.00019
Gauss-Jacobi Iteration 8: x1 = 0.999952, x2 = 0.999952
Gauss-Jacobi Iteration 9: x1 = 1.00001, x2 = 1.00002
Gauss-Jacobi Iteration 10: x1 = 0.999996, x2 = 0.999996
Gauss-Jacobi Iteration 11: x1 = 1, x2 = 1
Gauss-Jacobi Iteration 12: x1 = 1, x2 = 1
Gauss-Jacobi Iteration 13: x1 = 1, x2 = 1
Gauss-Jacobi Iteration 14: x1 = 1, x2 = 1
Gauss-Jacobi Iteration 15: x1 = 1, x2 = 1

======= GAUSS-SEIDEL ITERATIONS =======

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Seidel Iteration 1: x1 = 1.25, x2 = 0.916667
Gauss-Seidel Iteration 2: x1 = 1.02083, x2 = 0.993056
Gauss-Seidel Iteration 3: x1 = 1.00174, x2 = 0.999421
Gauss-Seidel Iteration 4: x1 = 1.00014, x2 = 0.999952
Gauss-Seidel Iteration 5: x1 = 1.00001, x2 = 0.999996
Gauss-Seidel Iteration 6: x1 = 1, x2 = 1
Gauss-Seidel Iteration 7: x1 = 1, x2 = 1
Gauss-Seidel Iteration 8: x1 = 1, x2 = 1
Gauss-Seidel Iteration 9: x1 = 1, x2 = 1
Gauss-Seidel Iteration 10: x1 = 1, x2 = 1
Gauss-Seidel Iteration 11: x1 = 1, x2 = 1
Gauss-Seidel Iteration 12: x1 = 1, x2 = 1
Gauss-Seidel Iteration 13: x1 = 1, x2 = 1
Gauss-Seidel Iteration 14: x1 = 1, x2 = 1
Gauss-Seidel Iteration 15: x1 = 1, x2 = 1

========== FINAL COMPARISON ==========

Methods                 No of iteration     Error %          Time (ms)        Status
-------------------------------------------------------------------------------
Jacobi method           8                   0.004800         48.484600        Tolerance reached
Gauss-Seidel method     5                   0.000700         46.510900        Tolerance reached
```

```
PS C:\Users\Mayank\Desktop\Gauss Jacobi and Seidel Iteration> ./test 2
Running Test Case 2 (n=4)

========== JACOBI ITERATIONS ==========

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Jacobi Iteration 1: x1 = 1.6, x2 = 1.26667, x3 = 1.3, x4 = 1.4
Gauss-Jacobi Iteration 2: x1 = 0.816667, x2 = 0.886667, x3 = 0.88, x4 = 0.856667
Gauss-Jacobi Iteration 3: x1 = 1.073, x2 = 1.03933, x3 = 1.0425, x4 = 1.053
Gauss-Jacobi Iteration 4: x1 = 0.974083, x2 = 0.985233, x3 = 0.98415, x4 = 0.980583
Gauss-Jacobi Iteration 5: x1 = 1.00965, x2 = 1.00537, x3 = 1.00578, x4 = 1.00713
Gauss-Jacobi Iteration 6: x1 = 0.996479, x2 = 0.998021, x3 = 0.997873, x4 = 0.997383
Gauss-Jacobi Iteration 7: x1 = 1.0013, x2 = 1.00073, x3 = 1.00078, x4 = 1.00096
Gauss-Jacobi Iteration 8: x1 = 0.999525, x2 = 0.999734, x3 = 0.999714, x4 = 0.999647
Gauss-Jacobi Iteration 9: x1 = 1.00017, x2 = 1.0001, x3 = 1.00011, x4 = 1.00013
Gauss-Jacobi Iteration 10: x1 = 0.999936, x2 = 0.999964, x3 = 0.999961, x4 = 0.999952
Gauss-Jacobi Iteration 11: x1 = 1.00002, x2 = 1.00001, x3 = 1.00001, x4 = 1.00002
Gauss-Jacobi Iteration 12: x1 = 0.999991, x2 = 0.999995, x3 = 0.999995, x4 = 0.999994
Gauss-Jacobi Iteration 13: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Jacobi Iteration 14: x1 = 0.999999, x2 = 0.999999, x3 = 0.999999, x4 = 0.999999
Gauss-Jacobi Iteration 15: x1 = 1, x2 = 1, x3 = 1, x4 = 1

======= GAUSS-SEIDEL ITERATIONS =======

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Seidel Iteration 1: x1 = 1.6, x2 = 1.16, x3 = 0.966, x4 = 0.9114
Gauss-Seidel Iteration 2: x1 = 0.98706, x2 = 1.01494, x3 = 1.00348, x4 = 0.997957
Gauss-Seidel Iteration 3: x1 = 0.996171, x2 = 1.0003, x3 = 1.00044, x4 = 1.00028
Gauss-Seidel Iteration 4: x1 = 0.999781, x2 = 0.999948, x3 = 1.00002, x4 = 1.00003
Gauss-Seidel Iteration 5: x1 = 1, x2 = 0.999995, x3 = 0.999999, x4 = 1
Gauss-Seidel Iteration 6: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 7: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 8: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 9: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 10: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 11: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 12: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 13: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 14: x1 = 1, x2 = 1, x3 = 1, x4 = 1
Gauss-Seidel Iteration 15: x1 = 1, x2 = 1, x3 = 1, x4 = 1

========== FINAL COMPARISON ==========

Methods                 No of iteration     Error %          Time (ms)        Status
-------------------------------------------------------------------------------
Jacobi method           11                  0.001500         35.686200        Tolerance reached
Gauss-Seidel method     5                   0.000150         37.547600        Tolerance reached
```

```
PS C:\Users\Mayank\Desktop\Gauss Jacobi and Seidel Iteration> ./test 3
Running Test Case 3 (n=5)

========== JACOBI ITERATIONS ==========

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Jacobi Iteration 1: x1 = 1.4, x2 = 1.4, x3 = 1.4, x4 = 1.4, x5 = 1.4
Gauss-Jacobi Iteration 2: x1 = 0.84, x2 = 0.84, x3 = 0.84, x4 = 0.84, x5 = 0.84
Gauss-Jacobi Iteration 3: x1 = 1.064, x2 = 1.064, x3 = 1.064, x4 = 1.064, x5 = 1.064
Gauss-Jacobi Iteration 4: x1 = 0.9744, x2 = 0.9744, x3 = 0.9744, x4 = 0.9744, x5 = 0.9744
Gauss-Jacobi Iteration 5: x1 = 1.01024, x2 = 1.01024, x3 = 1.01024, x4 = 1.01024, x5 = 1.01024
Gauss-Jacobi Iteration 6: x1 = 0.995904, x2 = 0.995904, x3 = 0.995904, x4 = 0.995904, x5 = 0.995904
Gauss-Jacobi Iteration 7: x1 = 1.00164, x2 = 1.00164, x3 = 1.00164, x4 = 1.00164, x5 = 1.00164
Gauss-Jacobi Iteration 8: x1 = 0.999345, x2 = 0.999345, x3 = 0.999345, x4 = 0.999345, x5 = 0.999345
Gauss-Jacobi Iteration 9: x1 = 1.00026, x2 = 1.00026, x3 = 1.00026, x4 = 1.00026, x5 = 1.00026
Gauss-Jacobi Iteration 10: x1 = 0.999895, x2 = 0.999895, x3 = 0.999895, x4 = 0.999895, x5 = 0.999895
Gauss-Jacobi Iteration 11: x1 = 1.00004, x2 = 1.00004, x3 = 1.00004, x4 = 1.00004, x5 = 1.00004
Gauss-Jacobi Iteration 12: x1 = 0.999983, x2 = 0.999983, x3 = 0.999983, x4 = 0.999983, x5 = 0.999983
Gauss-Jacobi Iteration 13: x1 = 1.00001, x2 = 1.00001, x3 = 1.00001, x4 = 1.00001, x5 = 1.00001
Gauss-Jacobi Iteration 14: x1 = 0.999997, x2 = 0.999997, x3 = 0.999997, x4 = 0.999997, x5 = 0.999997
Gauss-Jacobi Iteration 15: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1

======= GAUSS-SEIDEL ITERATIONS =======

Enter the size of the system (n): Enter the elements of matrix A (row by row):
Enter the elements of vector d (b vector):
Gauss-Seidel Iteration 1: x1 = 1.4, x2 = 1.26, x3 = 1.134, x4 = 1.0206, x5 = 0.91854
Gauss-Seidel Iteration 2: x1 = 0.966686, x2 = 0.996017, x3 = 1.00982, x4 = 1.01089, x5 = 1.00166
Gauss-Seidel Iteration 3: x1 = 0.998161, x2 = 0.997947, x3 = 0.999134, x4 = 1.00031, x5 = 1.00044
Gauss-Seidel Iteration 4: x1 = 1.00022, x2 = 0.999989, x3 = 0.999904, x4 = 0.999945, x5 = 0.999995
Gauss-Seidel Iteration 5: x1 = 1.00002, x2 = 1.00001, x3 = 1, x4 = 0.999997, x5 = 0.999997
Gauss-Seidel Iteration 6: x1 = 0.999999, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 7: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 8: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 9: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 10: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 11: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 12: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 13: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 14: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1
Gauss-Seidel Iteration 15: x1 = 1, x2 = 1, x3 = 1, x4 = 1, x5 = 1

========== FINAL COMPARISON ==========

Methods                 No of iteration     Error %          Time (ms)        Status
-------------------------------------------------------------------------------
Jacobi method           12                  0.001700         35.280200        Tolerance reached
Gauss-Seidel method     5                   0.000720         38.425600        Tolerance reached
```

CONCLUSION

Why Gauss-Seidel Converges Twice As Fast

About 2x

Convergence

Gauss-Seidel reaches identical tolerance in roughly half the iteration count across all systems.

Immediate Variable Updating

Newly computed components are immediately used in subsequent equations during the same iteration, allowing updated values to propagate faster and reducing the number of iterations required to achieve convergence compared to Jacobi.